Expansion versus Diversion in Global Green Markets: Evidence from Green Industrial Policy Announcements

Amanda Kennard* Justin Melnick†

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Abstract

Green industrial policies aim to accelerate decarbonization while increasingly favoring domestic producers. We distinguish between two mechanisms shaping their international consequences: market expansion, which benefits firms across borders, and market diversion, which shifts economic gains toward domestic firms at foreign competitors' expense. Using event studies of publicly traded clean energy firms in the United States, China, and Germany, we compare reactions to the European Union's REPowerEU Plan and the U.S. Inflation Reduction Act (IRA). REPowerEU generated significant gains among Chinese firms, particularly those with greater foreign sales exposure. The IRA increased valuations for U.S. firms but generated little response among Chinese firms. However, proposed reductions in IRA support subsequently produced losses among U.S. firms and gains among Chinese firms, consistent with the effects of market diversion becoming increasingly certain over time. These findings highlight how preferentialism shapes the international distributional consequences of green industrial policy.

*Department of Political Science, Stanford University, amanda.kennard@stanford.edu
†Niehaus Center for Globalization and Governance, Princeton University, melnickj@princeton.edu

Climate policy is widely implemented through instruments of "green industrial policy" (Rodrik 2014; Nemet 2019; Allan, Lewis and Oatley 2021; Allan and Nahm 2025). Across major economies, governments have adopted subsidies, tax credits, and public investment programs designed to support the growth of clean energy industries and accelerate decarbonization (Meckling 2021; 2025). Yet these policies increasingly pursue a parallel objective: ensuring that domestic, rather than foreign, firms capture the economic benefits generated by the clean energy transition. The European Union's REPowerEU plan and the United States' Inflation Reduction Act are among the most consequential examples of this approach, combining unprecedented support for clean energy investment with provisions designed to channel the resulting economic gains toward domestic industries. This strategy reflects a growing belief that climate policy is more likely to endure when its economic benefits are concentrated among politically salient domestic constituencies (Meckling et al. 2015).

This combination of market expansion and preferential support for domestic firms creates a tension in the design of contemporary green industrial policy: if domestic producers gain from these policies, does this reflect aggregate growth in global markets or a reallocation of productive activity across national borders? We develop a framework that distinguishes between market expansion, in which industrial policy enlarges global markets for clean technologies, and market diversion, in which it redirects production and market share toward domestic firms at the expense of foreign competitors. These mechanisms are not mutually exclusive, and whether the international effects of any given policy are dominated by market expansion or market diversion is theoretically ambiguous. Yet understanding which effects dominate is increasingly important as governments rely more heavily on industrial policy to accelerate the clean energy transition.

We examine this question by comparing the international effects of two major green industrial policy initiatives: the REPowerEU Plan in the European Union and the Inflation Reduction Act (IRA) in the United States. Both initiatives have the potential

to generate market expansion and market diversion, but differ in the relative emphasis placed on expanding demand for clean technologies versus promoting domestic production. The IRA represents the largest expansion of U.S. support for clean energy industries in history, providing hundreds of billions of dollars in subsidies and tax incentives for renewable energy, batteries, electric vehicles, and related technologies (Gazmararian, Jensen and Tingley 2026). Many of these incentives are coupled with preferential provisions, including domestic content bonuses and measures designed to direct the gains from market expansion toward U.S. firms (Leonelli and Clora 2024). By contrast, REPowerEU focuses primarily on accelerating the deployment of renewable energy and reducing dependence on imported fossil fuels, with comparatively fewer provisions explicitly favoring European producers.[1]

The comparison between these initiatives provides an opportunity to examine how preferentialism shapes the international consequences of green industrial policy. Under the logic of market expansion, policies that stimulate demand for clean technologies should benefit firms both within and outside the implementing jurisdiction. Green industrial policy not only encourages technological adoption by reducing costs for consumers; it also supports investment and production across clean energy supply chains, enabling firms to reduce costs by achieving economies of scale or pursuing new innovation. These effects can generate positive externalities across interconnected production networks, reinforcing the benefits of expanding demand. Where market expansion dominates, these gains are likely to extend across national borders, benefiting foreign firms both through increased sales and through the spillovers generated by expanding industrial ecosystems. Because REPowerEU places comparatively less emphasis on preferential support for domestic producers, we expect its international effects to be characterized primarily by

[1]Recent work by international organizations has highlighted the international competitive effects of industrial policy, including its implications for trade, investment, and competitiveness across countries (Hodge 2024; Huang et al. 2025). Our contribution, by contrast, is to distinguish between expansionary and diversionary channels within green industrial policy and to examine how their relative importance varies across policy settings.

market expansion.

Under the logic of market diversion, by contrast, preferential provisions alter the international distribution of these benefits. By directing investment and production toward the domestic economy, these policies aim to concentrate gains at home, limiting the extent to which foreign competitors benefit from market expansion. Moreover, as with market expansion, the resulting competitive advantages may extend beyond the domestic market. By enabling domestic producers to expand production, realize economies of scale, and strengthen their supply chains, preferential provisions may improve their competitiveness in global markets, potentially disadvantaging foreign competitors there as well. Strengthening the international competitiveness of U.S. clean energy producers was an explicit objective of the Biden administration's approach to green industrial policy (The White House 2022; 2023). The IRA's preferential provisions thus yield two potential implications for foreign firms. If market diversion merely offsets the benefits of market expansion, foreign firms may experience little or no net gain from the policy. If market diversion dominates market expansion, however, foreign firms may experience outright losses.

We evaluate these predictions using an event study of publicly traded clean energy firms in the United States, China, and Germany.[2] Specifically, we examine stock market reactions to two major announcements: the European Commission's release of the REPowerEU Plan on May 18, 2022, and the announcement of the Schumer-Manchin agreement which paved the way for the Inflation Reduction Act on July 27, 2022.[3] Both announcements represented substantial positive shocks to expectations about future government support for clean energy industries, but differed in the extent to which the pro-

[2]Our empirical design is similar in spirit to Aklin (2018) which studies the impacts of the 2016 election on clean energy firms globally.

[3]We focus on the release of the detailed REPowerEU Plan in May 2022 rather than the initial announcement in March, which coincided with substantial geopolitical and market uncertainty following Russia's invasion of Ukraine. The May release also provides a more detailed account of the proposed policy measures. Bayer, Crippa and Genovese (2026) examines the broader development of REPowerEU and its effects on European clean energy and fossil fuel firms.

posed policies were expected to incorporate preferential provisions. Our sample comprises approximately 130 firms involved in the manufacturing of clean energy technologies and their components, identified using Bloomberg and Refinitiv industry classifications. We estimate abnormal stock returns around each announcement using country-specific factor models to account for broader market movements (Fama and French 1993).

Our results reveal contrasting patterns in the international effects of REPowerEU and the IRA. Consistent with the logic of market expansion, the announcement of REPowerEU generated positive abnormal returns among clean energy firms outside the European Union. Chinese firms experienced statistically significant gains of approximately 3 percentage points, while U.S. firms also registered positive returns, although these estimates are noisy and not statistically significant. Among Chinese firms, gains were significantly larger for those with greater exposure to foreign markets, consistent with the benefits of expected growth in foreign demand. However, firms with relatively low foreign sales also experienced significant positive returns, suggesting that the benefits of market expansion may also propagate into domestic supply chains and production networks. By contrast, the announcement of the IRA generated substantial positive abnormal returns among U.S. clean energy firms—averaging approximately 13 percentage points—but little response among Chinese firms and imprecisely estimated effects among German firms. These findings suggest that the anticipated benefits of REPowerEU extended beyond the implementing jurisdiction, whereas those of the IRA were concentrated primarily within the domestic economy.

One possible explanation for the absence of a significant response among Chinese firms to the initial IRA announcement is that investors remained uncertain about the extent to which its preferential provisions would ultimately benefit U.S. producers at the expense of foreign competitors. To explore this possibility, we also estimate abnormal returns following the June 2025 announcement of proposed reductions in IRA support under the One Big Beautiful Bill Act (OBBBA). We find substantial negative abnormal

returns among U.S. clean energy firms, averaging approximately 4.5 percentage points, alongside significant positive returns of approximately 1.5 to 2 percentage points among Chinese firms. These findings are consistent with the logic of market diversion, suggesting that U.S. industrial support strengthened the competitive position of domestic producers relative to foreign firms, though the magnitude and extent of these distributional effects may have remained uncertain to investors at the time of the IRA's initial announcement. The results are robust to a range of estimation windows, inverse-variance weighting, and an alternative LASSO-based approach to counterfactual estimation.

Our analysis contributes to the growing political economy literature on green industrial policy by shifting attention from its domestic foundations to its international consequences. Existing work has emphasized the role of industrial policy in accelerating decarbonization and building domestic political support for climate action (Meckling 2021; Gazmararian, Jensen and Tingley 2026). In contrast, scholarship on the international effects of climate policy has remained largely agnostic about the specific instruments through which climate policy is implemented (Genovese 2019; Kennard 2020; Colgan, Green and Hale 2021). We bridge these literatures by distinguishing two mechanisms through which green industrial policy affects foreign firms: market expansion, which generates benefits across national borders, and market diversion, which redistributes economic gains toward domestic firms at the expense of foreign competitors. Our analysis further highlights the importance of preferentialism in shaping the balance between these mechanisms. By distinguishing between the expansion of global markets and the international redistribution of economic activity, we provide a framework for understanding why the international consequences of green industrial policy may vary substantially across policy settings (Melnick 2026).

More broadly, our results highlight a potential tension embedded in the political logic of green industrial policy. Existing scholarship emphasizes the potential of green industrial policy to cultivate domestic political constituencies that favor pro-climate reform

(Meckling et al. 2015; Meckling 2021). This builds on a growing emphasis within the field on the importance of the domestic distributional foundations of successful climate policy (Aklin and Mildenberger 2020, though see also Kennard and Schnakenberg (2023) and Rowan (2025)). Yet the international consequences of policies designed to build these constituencies depend on how their economic benefits are distributed. Where market expansion dominates, green industrial policy may strengthen clean energy industries beyond the implementing jurisdiction, potentially reinforcing the economic interests that support decarbonization abroad. Where preferential provisions concentrate these benefits among domestic producers, however, the same policies may undermine the competitiveness and profitability of foreign clean energy firms. Strategies designed to build durable domestic constituencies for climate action may therefore generate positive international spillovers, but they may also weaken the economic interests that could support decarbonization in other jurisdictions.

1 International Effects of Green Industrial Policy

Green industrial policy has become an increasingly central feature of national climate strategies. Although governments have long used industrial policy to shape economic development, what distinguishes the current moment is the scale, ambition, and political prominence with which these tools are being deployed in pursuit of decarbonization. Green industrial policy relies on state intervention to direct economic activity toward the development and eventual deployment of clean technologies through instruments such as subsidies, tax credits, and concessional finance. This transformation is evident in the proliferation of major policy initiatives, from the U.S. Inflation Reduction Act and the European Union's Green Deal to comparable programs in China, Japan, South Korea, and across the Global South (Rodrik 2014; Meckling 2021).

This resurgence of green industrial policy reflects both economic and political consid-

erations. Economically, industrial policy is often justified as a response to market failures that impede investment in emerging clean technologies. Politically, however, it offers governments a means of building durable domestic support for climate action by creating jobs, attracting investment, and cultivating industrial constituencies with a stake in the energy transition. Unlike traditional climate instruments, which primarily seek to internalize the social costs of greenhouse gas emissions, green industrial policy also aims to generate visible economic benefits that strengthen political support for continued decarbonization (Meckling, Sterner and Wagner 2017; Stern and Stiglitz 2023).

A central objective of green industrial policy is to expand markets for low-carbon technologies. In the absence of state support, firms and consumers may underinvest in these technologies because of coordination failures, financing constraints, or a failure to internalize the long-run social benefits of early investment. By reducing barriers to investment, industrial policy accelerates the development, production, and deployment of clean technologies while stimulating additional private investment and innovation. As production expands, learning-by-doing and economies of scale reduce costs, encouraging broader adoption and reinforcing market growth. The result is an expansion in the scale of clean technology markets, supporting the diffusion of low-carbon alternatives throughout the economy (Rodrik 2014; Melnick 2026).

A classic illustration is the global solar photovoltaic industry during the 2000s. Germany's generous feed-in tariffs created a large and predictable market for solar panels, helping to accelerate deployment, reduce costs, and stimulate investment throughout the global supply chain. Yet many of the resulting production and learning effects occurred outside of Germany. Chinese manufacturers rapidly expanded capacity, benefiting from scale economies and learning-by-doing, and soon emerged as dominant global producers. The German case therefore illustrates both the promise and the political vulnerability of market-expanding green industrial policy: demand-side support can accelerate global technological change, but the industrial gains from that expansion may accrue elsewhere

(Nemet 2019; Nahm 2021).

This experience helps explain why contemporary green industrial policies have increasingly incorporated measures designed to influence the location of green production. Rather than simply accelerating the deployment of clean technologies, governments increasingly seek to shape where those technologies are produced and how global green supply chains are organized. Recent policies therefore extend beyond technology-neutral support for decarbonization to include provisions explicitly intended to encourage production and investment at home. These measures include domestic content requirements, local manufacturing incentives, production subsidies tied to domestic facilities, and tax credits contingent on local sourcing or assembly. As a result, contemporary green industrial policy increasingly combines broad support for clean technologies with targeted incentives designed to influence the location of production (Mazzocco 2022; Meckling 2025).[4]

The growing use of preferential provisions therefore creates a tension in the design of contemporary green industrial policy. On the one hand, industrial policy seeks to accelerate investment and expand global markets for clean technologies by overcoming market failures. On the other hand, preferential provisions may generate beggar-thy-neighbor effects by diverting production, investment, and market share toward domestic firms at the expense of foreign competitors. We refer to these competing channels as market expansion and market diversion, respectively.

This framework parallels the well-established distinction between business expansion and business stealing in the industrial organization literature (Mankiw and Whinston 1986; Tirole 1988), and also bears some resemblance to the classic distinction between

[4]Foreign firms may also benefit from these incentives by relocating or expanding production within the implementing jurisdiction. Yet doing so requires costly investment and is frequently constrained by ownership restrictions or other regulatory obstacles. Importantly, such responses may mitigate the distributional consequences of preferential policies without eliminating the underlying incentives to redirect production and investment toward the domestic economy. We consider this possibility in our empirical analysis below.

trade creation and trade diversion in the international trade literature (Viner 1950). Firms can increase profits either by expanding the overall size of a market or by capturing market share from competing firms. An analogous distinction applies to green industrial policy. Rather than maximizing firm profits, governments seek to increase domestic investment, employment, and industrial capacity. They can do so either by expanding the overall market for clean technologies or by redirecting production and market share toward domestic firms. The central question is not simply whether green industrial policy benefits domestic producers, but whether it does so primarily by enlarging global markets or by redistributing pre-existing economic activity across countries.

While both market expansion and market diversion are likely to coexist in many green industrial initiatives, their relative importance may vary considerably across policy contexts. Initiatives that primarily seek to accelerate the deployment of clean technologies may generate substantial market-expanding effects, while those that incorporate preferential incentives for domestic production may be more likely to divert investment and market share from foreign competitors. Nevertheless, the balance between these mechanisms cannot be inferred from policy design alone. Their ultimate effects depend on national economic circumstances, the organization of global supply chains, and the responses of firms, investors, and consumers (Mazzocco 2022; Meckling 2025). Distinguishing which mechanism dominates therefore requires an empirical evaluation of how investors respond to green industrial policy initiatives with differing objectives and institutional designs.

2 Empirical Setting and Predictions

We test competing implications of market expansion versus market diversion in two parallel contexts: the 2022 U.S. Inflation Reduction Act and the European Union's RE-PowerEU initiative. Both represent major green industrial policy interventions aimed at accelerating the energy transition through large-scale investments into clean energy

technologies. The IRA provides an estimated $370 billion in federal support for clean energy, transportation, manufacturing, and related technologies through a combination of tax credits, grants, and other incentives (Gazmararian, Jensen and Tingley 2026). REPowerEU, introduced in response to Russia's invasion of Ukraine, similarly mobilized substantial resources to accelerate renewable energy deployment, expand clean energy infrastructure, and reduce Europe's dependence on Russian fossil fuels. The European Commission estimated that achieving these objectives would require approximately €210 billion in additional investment by 2027 (European Commission 2022). Given the scale of these initiatives and the size of the U.S. and European economies, both have the potential to influence investment decisions, supply chains, and competitive dynamics well beyond their respective borders (International Energy Agency 2024; Bermel et al. 2024).

Beyond considerations of scale, both the IRA and REPowerEU exemplify the broader resurgence of green industrial policy described above. Rather than relying primarily on carbon pricing or environmental regulation, both initiatives seek to accelerate decarbonization through public investment and incentives for the production and deployment of clean technologies. They therefore place state intervention at the center of climate strategy while seeking to mobilize private investment in strategically important industries. As a result, the IRA and REPowerEU represent prominent examples of the renewed role of industrial policy in climate governance and have become focal points in debates over the state's role in shaping the energy transition (Bistline and Wolfram 2025; Leonelli and Clora 2024; European Commission 2022).

The IRA and REPowerEU differ, however, in the extent to which they incorporate preferential provisions designed to benefit domestic industries. While the IRA seeks to accelerate the deployment of clean technologies, many of its benefits are explicitly tied to domestic production, manufacturing, or sourcing. Tax credits for electric vehicles, batteries, and renewable energy technologies incorporate domestic content requirements or provide additional incentives for goods produced in the United States. These provisions re-

flect the Biden administration's broader objectives of rebuilding domestic manufacturing, creating high-quality employment, and strengthening strategically important industries alongside decarbonization (Leonelli and Clora 2024). By contrast, REPowerEU places greater emphasis on accelerating the deployment of renewable energy and expanding clean energy infrastructure, with fewer provisions explicitly favoring European manufacturers. Although the initiative also seeks to strengthen European industrial capacity, its principal instruments are directed toward expanding investment and demand for clean technologies rather than redirecting production toward domestic firms (European Commission 2022).

These differences make the IRA and REPowerEU useful cases for evaluating our theoretical framework. Both initiatives have the potential to generate market expansion and market diversion, but differ in the relative emphasis placed on expanding demand for clean technologies versus promoting domestic production. The IRA's preferential provisions suggest a greater potential for market diversion, while REPowerEU's emphasis on accelerating deployment suggests a stronger role for market expansion. The relative importance of these mechanisms in shaping investor expectations is therefore an empirical question. We address this question by examining how the announcements of these initiatives affect investor valuations of clean energy firms in the United States, Germany, and China.

Our framework yields distinct empirical implications for the aggregate and international distributional effects of green industrial policy. Under the logic of market expansion, policies that stimulate demand, investment, and innovation in clean technologies should increase the expected value of firms across the global clean energy sector. We therefore expect market expansion to generate positive aggregate abnormal returns, with gains extending to firms both within and outside the adopting jurisdiction. Market diversion, by contrast, redistributes expected economic benefits toward domestic producers at the expense of foreign competitors, without itself increasing aggregate industry value. Preferential provisions should therefore benefit domestic firms relative to foreign competitors.

Because the two mechanisms can operate simultaneously, however, the implications for foreign firms depend on their relative strength. If market diversion partially offsets the benefits of market expansion, foreign firms may experience smaller positive returns or little net change in valuation; if diversion dominates, they may experience outright losses. These distributional effects are particularly relevant for Chinese firms, which are major competitors in global clean technology markets (Mazzocco 2022).

Given the greater emphasis on preferential support for domestic production under the IRA, we expect market diversion to play a more prominent role in the response to the U.S. announcement. The IRA may therefore generate positive aggregate returns while concentrating gains among U.S. firms and producing smaller or even negative returns among foreign competitors. Conversely, REPowerEU's emphasis on expanding renewable energy deployment suggests that market expansion should dominate, generating positive aggregate returns and gains extending to firms outside the European Union.

3 Research Design

We employ an event study approach to evaluate how investors revised their valuations of clean energy manufacturers in the United States, China, and Germany in response to both the European Commission's REPowerEU plan and the agreement that produced the Inflation Reduction Act. Comparing abnormal returns across countries, and across firms within China, allows us to assess whether investors expected each policy to expand the global market, to shift market share toward domestic firms, or both.[5]

[5]Event studies are widely used in political science and economics to examine how investors respond to political developments (e.g., Wilf 2016). In the context of climate politics, see Aklin (2018); Bayer, Crippa and Genovese (2026); Martins et al. (2025); Voeten (2025); Bauer, Offner and Rudebusch (2026).

3.1 Events and Their Informational Content

We examine two major announcements concerning the expansion of government support for clean energy industries: the European Commission's release of the REPowerEU Plan on May 18, 2022, and the Schumer-Manchin agreement on the Inflation Reduction Act on July 27, 2022. Because stock prices incorporate expectations about future policy, abnormal returns around these announcements capture revisions in investor expectations relative to the information already available. Our estimates therefore measure the market response to the new information revealed by each announcement, rather than the full valuation effects of the policies relative to a counterfactual in which they were never adopted.

The REPowerEU Plan was presented by the European Commission on May 18, 2022. Building on the initial REPowerEU announcement in March, the Commission released a detailed package of policy proposals intended to accelerate renewable energy deployment across the European Union. These included raising the EU's proposed 2030 renewable energy target from 40 to 45 percent, establishing a target of nearly 600 GW of installed solar capacity by 2030, and introducing measures to accelerate permitting and expand rooftop solar deployment (European Commission 2022). The release provided a more detailed account of the regulatory, financial, and implementation measures through which the Commission intended to achieve these targets, potentially increasing the credibility of its commitments to expanding clean energy deployment.

Drafts of the plan, including its proposed solar targets, had circulated in the weeks preceding its release (Bayer, Crippa and Genovese 2026). Investors may therefore have incorporated some of the anticipated expansion in European clean energy support into firm valuations before the event window. Our estimates capture the additional information conveyed by the formal release of these proposals, including any revision to its perceived credibility and likelihood of implementation, relative to this *ex ante* baseline. The plan

was released during European trading hours, after the close of Chinese markets. We therefore set the event date ($t = 0$) to May 18 for firms listed in Germany and the United States and to May 19 for firms listed in China.

The informational context surrounding the IRA announcement differed considerably. The Schumer-Manchin agreement was announced in the late afternoon of July 27, 2022, following the failure of the Biden administration's earlier Build Back Better proposal and significant uncertainty about the prospects for federal climate legislation. Two weeks earlier, Manchin had signaled that he would not support new climate and energy spending, which was widely understood to extinguish any remaining hope for climate legislation in that Congress. The agreement was therefore unanticipated and represented a substantial positive shock to expectations about the likelihood of federal support for clean energy industries. At the same time, given the context investors may have remained uncertain about the ultimate form of the legislation's implementation and the final distribution of its economic benefits. Because the announcement came after U.S. markets had closed and before Asian and European markets opened, we set $t = 0$ to July 28, 2022, in all three markets. We use the agreement rather than the bill's final passage or signature in August because passage was widely anticipated once the agreement was announced.

These differences in informational content matter for the interpretation of our results. Differences in the magnitude of abnormal returns across the two announcements may reflect differences in prior expectations and the size of the information shocks, as well as differences in policy design. We therefore do not interpret the comparison as isolating the causal effect of preferential provisions. Instead, we examine whether the pattern of responses among domestic and foreign firms is consistent with the mechanisms of market expansion and market diversion developed in our theoretical framework.

3.2 Sample

We identify clean energy manufacturing firms using Bloomberg's industry classifications, supplemented by the LSEG/Refinitiv Business Classification system. For firms covered by Bloomberg, we use the Bloomberg Industry Classification System segment where available and the Global Industry Classification Standard sub-industry otherwise. The resulting sample spans the solar photovoltaic supply chain, wind turbines and components, energy storage, grid equipment, fuel cells, and electric vehicles. We exclude electricity generators identified as such by both classification systems, whose valuations depend primarily on power markets, as well as fossil fuel and biofuel producers and firms for which clean energy manufacturing is incidental to their operations. Although the sample consists primarily of equipment manufacturers, it also includes some firms engaged in clean energy project development and installation.

Firms are assigned to countries by headquarters. U.S.-listed shares of Chinese firms (e.g., JinkoSolar, Daqo New Energy) are therefore grouped with Chinese firms, but their expected returns are modeled using factors for the U.S. market, where the shares trade. The resulting samples include 131 firms for REPowerEU (85 in China, 34 in the United States, and 12 in Germany) and 134 firms for the IRA (88, 34, and 12, respectively). Appendix A1 details the classification rules and firm-level exclusions.

3.3 Data

We obtain daily prices from Compustat Global and Compustat North America via Wharton Research Data Services. We adjust prices for splits and other capital changes using Compustat's daily adjustment factor and for dividends using its total return factor, which is available for all Chinese and German securities and for a minority of U.S. securities; for the remaining U.S. securities, returns exclude dividends. We convert prices to U.S. dollars using Compustat daily exchange rates, compute simple daily returns, and subtract

the daily U.S. one-month Treasury bill rate from Kenneth French's data library to obtain excess returns. We drop days on which the exchange where a security trades is closed, so that returns after a holiday are measured from the last trading day before it.

To examine the channels underlying the country-level estimates, we also compare Chinese firms that differ in their exposure to foreign markets. We measure each firm's share of sales outside its home country using Worldscope, drawing on the last fiscal year ending before each event (fiscal year 2021 for both events, or the closest earlier year back to 2019 if 2021 is unavailable). We classify Chinese firms as having high or low foreign exposure depending on whether their foreign sales share falls above or below the sample median. Under the logic of market expansion, firms with greater exposure to foreign markets should benefit more from policies that stimulate demand abroad.

3.4 Estimation

We model each firm's excess returns using a three-factor model with daily, country-specific market, size, and value factors from the Global Factor Data of Jensen, Kelly and Pedersen (2023). We use the market factor and the size (market equity) and value (book-to-market) factors for the country in which each security trades, in their capped value-weighted versions. The factors are expressed as U.S. dollar returns in excess of the U.S. Treasury bill rate, so they are measured in the same currency and at the same frequency as our firm returns. For firm i listed in country $c(i)$ on day t:

$$R_{it} - R_t^f = \alpha_i + \beta_{1i}\, MKT_{c(i)t} + \beta_{2i}\, SMB_{c(i)t} + \beta_{3i}\, HML_{c(i)t} + \varepsilon_{it},$$

where $MKT_{c(i)t}$ is the market excess return, $SMB_{c(i)t}$ the size factor, and $HML_{c(i)t}$ the value factor, measured at the country level on each trading day. We estimate the model by ordinary least squares separately for each firm over an estimation window that ends five trading days before the event. The abnormal return on day t of the event window is

the realized excess return minus the model's prediction,

$$AR_{it} = (R_{it} - R_t^f) - \left(\hat{\alpha}_i + \hat{\beta}_{1i}\, MKT_{c(i)t} + \hat{\beta}_{2i}\, SMB_{c(i)t} + \hat{\beta}_{3i}\, HML_{c(i)t}\right),$$

and the firm's cumulative abnormal return over the two-day event window is $CAR_i = AR_{i0} + AR_{i1}$. Day 0 is the first trading session in which the news could be priced (for REPowerEU in Germany and the United States, the session during which the plan was released), and the two-day window adds the following session. This choice is standard in event studies with staggered information release (MacKinlay 1997; Kothari and Warner 2007).

We estimate each firm's factor model over a window that ends five trading days before the event, and we report results for windows of 90, 60, and 30 trading days. The estimation window is intended to capture each firm's normal relationship to the market. Thus major news impacting our firms of interest during the estimation window is likely to distort both the estimated factor loadings and the variance estimate which our tests rely on. Selecting an appropriate window length therefore involves a trade-off in our setting: longer windows make use of more data but reach further back into major events such as Russia's invasion of Ukraine in February 2022 as well as the announcement of resulting U.S. sanctions. To assess this trade-off, we estimate the factor model employing windows of 30 to 180 trading days and score each version on the same held-out days surrounding each event (days -4 to -1 and $+2$ to $+10$).

The results are presented in Appendix Table A-2. Increasing the estimation window beyond 90 days yields almost no improvement: the median firm's prediction error is around one percent of the 90-day error and longer windows predict better for around half of firms in the sample. Windows shorter than 90 days perform slightly less well: the median firm's prediction error is at most 2 percent higher with a 60-day window and at most 5 percent higher with a 30-day window. We therefore focus on 90, 60, and 30 day

windows. These reduce contamination from other shocks without meaningfully sacrificing performance. Appendix Figure A-1 plots average predicted versus observed returns for the 90 day window.[6]

Our main quantities are the average cumulative abnormal returns of firms in each country and across firms. Because all firms share the same event date, abnormal returns will be correlated: a common shock to clean energy stocks on the event day moves many firms at once. Test statistics that treat firms as independent observations overstate precision (Kolari and Pynnönen 2010). We therefore base inference on the portfolio approach of Brown and Warner (1985).[7] Comparisons between groups—such as Chinese firms with high and low exposure to foreign markets—apply the same procedure to the daily difference between the two groups' average abnormal returns.

3.5 Identifying Assumptions

Our interpretation of abnormal returns as responses to the policy announcements assumes that no other news disproportionately affecting clean energy firms coincided with the event windows. Because expected returns are modeled relative to each firm's national market, our approach accounts for broader market movements on the event days. However, contemporaneous news affecting clean energy firms differently from the broader market could still confound our estimates. We therefore interpret abnormal returns as responses to the policy announcements conditional on the absence of such competing information shocks.

We also assume that firms' headquarters provide a reasonable proxy for their expo-

[6] We also describe results using the 120, 150, and 180 day windows in Section 4.1 below.

[7] For each group g (either at the country or pooled level) we form the daily average abnormal return $\overline{AR}_{gt} = 1/N_{gt} \sum_{i \in g} AR_{it}$ across the firms observed on day t, where abnormal returns in the estimation window are the residuals of each firm's factor model. The group's cumulative average abnormal return is $CAAR_g = \overline{AR}_{g0} + \overline{AR}_{g1}$, which equals the average of the firms' CAR_i. Its standard error is $\hat{\sigma}_g\sqrt{2}$, where $\hat{\sigma}_g$ is the standard deviation of $\overline{AR}_{gt}$ over the estimation window, and we compare $CAAR_g/(\hat{\sigma}_g\sqrt{2})$ with a t distribution with degrees of freedom equal to the number of estimation-window days minus one. Because $\hat{\sigma}_g$ is estimated from the time series of the group average, it incorporates the correlation of abnormal returns across firms.

sure to the preferential provisions of the policies we study. This assumption is potentially complicated by multinational production, since foreign-headquartered firms may also operate manufacturing facilities within the implementing jurisdiction. For Chinese firms, we assess the extent of this complication using the Clean Investment Monitor, which records clean energy manufacturing facilities announced in the United States since 2018. We identify Chinese firms in our sample that had announced or were operating a U.S. facility by the date of the Schumer-Manchin agreement. Only 3 of the 88 Chinese firms in our sample met this criterion, suggesting that headquarters is a reasonable proxy for exposure to U.S. domestic production preferences in this setting.

We note that the excess returns estimated via this procedure identify the realized distributional consequences of policy announcements only if investors price them correctly. We therefore interpret our estimates as evidence about where investors expected the gains from green industrial policy to accrue, not as measures of realized benefits.

4 Results

Figure 1 presents the two-day cumulative abnormal returns of clean energy firms following the announcements of REPowerEU and the Schumer-Manchin agreement on the IRA.[8] Both announcements generated positive aggregate abnormal returns, but the distribution of gains across countries differed substantially. Following REPowerEU, Chinese firms experienced statistically significant gains of approximately 3 percentage points, while U.S. firms also registered positive returns, although these estimates were less precise. German firms experienced smaller, statistically insignificant returns.

The announcement of the IRA, by contrast, generated substantial positive abnormal returns among U.S. firms, averaging approximately 13 percentage points, but little response among Chinese firms and no statistically significant response among German

[8]Tables corresponding to all results figures can be found in the Appendix.

firms. These patterns are robust to alternative estimation windows. Notably, across both events and all estimation windows, estimated returns for German firms remain close to zero, with consistently wide confidence intervals. Given the small number of German firms in our sample ($N = 12$), these estimates are insufficiently precise to distinguish between substantively meaningful positive and negative effects. We therefore interpret the German results with caution rather than as evidence of an absence of market effects.

Taken together, the results are consistent with some degree of market expansion following both announcements.[9] REPowerEU generated positive aggregate returns, including significant gains among foreign firms, whereas the IRA's positive aggregate returns were driven primarily by gains among U.S. producers. The contrasting distribution of these gains is also consistent with market diversion: unlike REPowerEU, the IRA generated little response among Chinese firms, suggesting that the benefits of market expansion may have been offset by the anticipated effects of preferential provisions. At the same time, investors may not yet have fully anticipated the distributional consequences of these provisions at the time of the initial announcement. We examine this possibility below by considering how firms responded to subsequent policy developments, including the anticipated withdrawal of IRA support.

Figure 2 examines heterogeneity in Chinese firms' responses according to their exposure to foreign markets. Following the announcement of REPowerEU, firms with greater foreign sales exposure experienced abnormal returns of approximately 4 percentage points, compared with approximately 2 percentage points among less internationally oriented firms. This difference is statistically significant, consistent with the expectation that firms more exposed to international markets benefit disproportionately from an expansion in foreign demand. Importantly, however, firms with lower foreign sales shares also experienced positive and statistically significant abnormal returns. This suggests that the

[9] We interpret positive aggregate abnormal returns as consistent with market expansion, but note that our estimates are equally weighted averages of firm-level percentage returns rather than changes in total industry valuation.

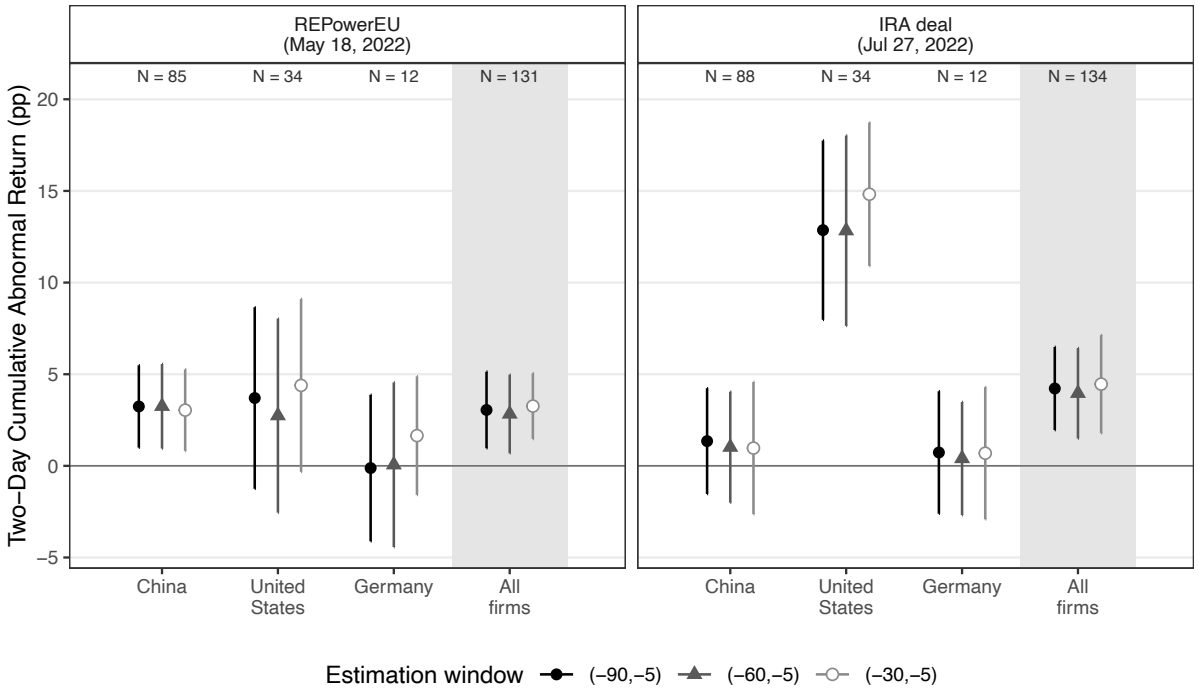


Figure 1: Abnormal Returns After Green Industrial Policy Announcement
Note: Mean country-level abnormal returns with 95% confidence intervals using portfolio test standard errors. Firm-level abnormal returns obtained with a three-factor model.

anticipated benefits of market expansion extend beyond firms directly exposed to foreign markets, potentially reflecting broader spillovers through domestic supply chains, production networks, and industry-wide growth expectations. By contrast, the announcement of the IRA generated smaller and statistically insignificant returns for both groups, with no meaningful differences by foreign sales exposure.

At the time of its announcement, the IRA combined substantial incentives for clean technology deployment with preferential provisions intended to support domestic manufacturing. However, the extent to which these provisions would ultimately favor U.S. producers over foreign competitors remained uncertain, depending in part on subsequent implementation guidance and investment decisions. To investigate this possibility, we examine investor responses to the June 16, 2025 announcement of the Senate Finance Committee's draft of the One Big Beautiful Bill Act, which proposed substantial reduc-

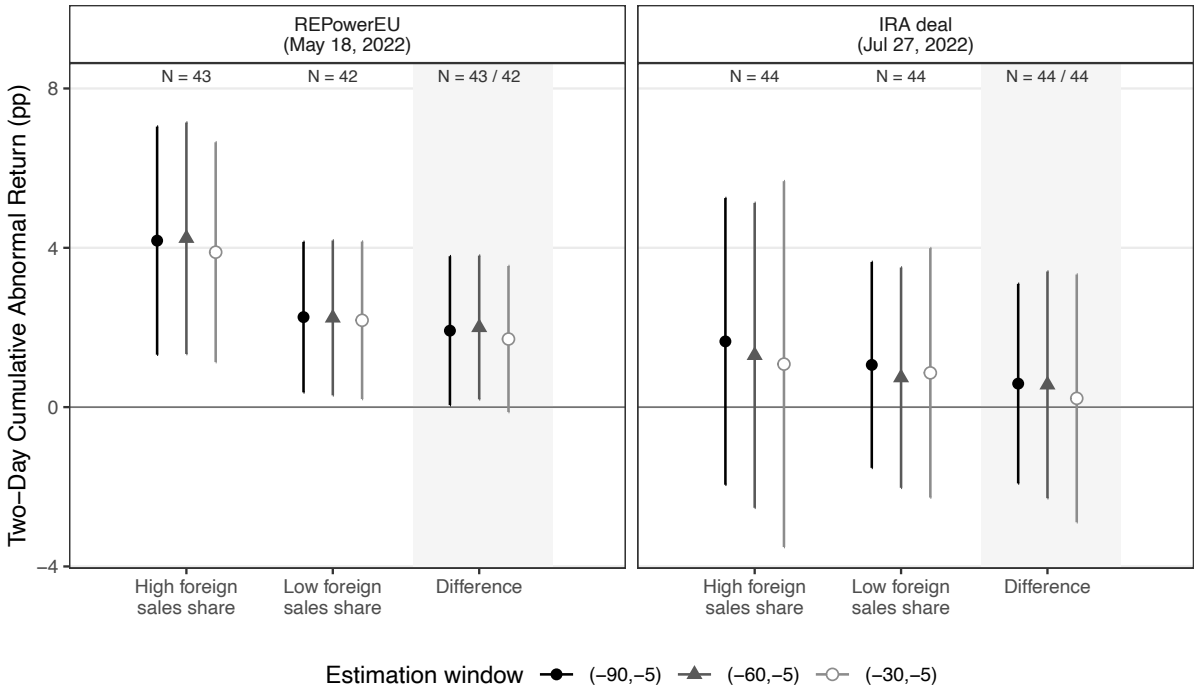


Figure 2: Abnormal Returns to Chinese Firms by Foreign Sales Share

Note: Mean abnormal returns for Chinese firms with high and low exposure to foreign markets with 95% confidence intervals using portfolio test standard errors. Firm-level abnormal returns obtained with a three-factor model.

tions in the clean energy incentives established under the IRA. Among other provisions, the draft accelerated the phaseout of production and investment tax credits for wind and solar energy, proposed eliminating federal electric vehicle purchase credits, and curtailed certain manufacturing incentives (Schurle and Roessler 2025). By this point, investors had considerably more information about the implementation of the IRA and the competitive implications of its preferential provisions. The proposed rollback therefore provides an additional opportunity to assess whether investors anticipated distributional consequences that were not apparent at the legislation's initial announcement.[10]

We employ the same estimation strategy used to examine the effects of REPowerEU and the IRA to study the effects of the OBBBA Senate draft on clean energy firms. Figure 3 presents the two-day cumulative abnormal returns following the announcement of

[10]The draft was released after the closure of U.S. markets on June 16, 20205, so day 0 is June 17 in all markets.

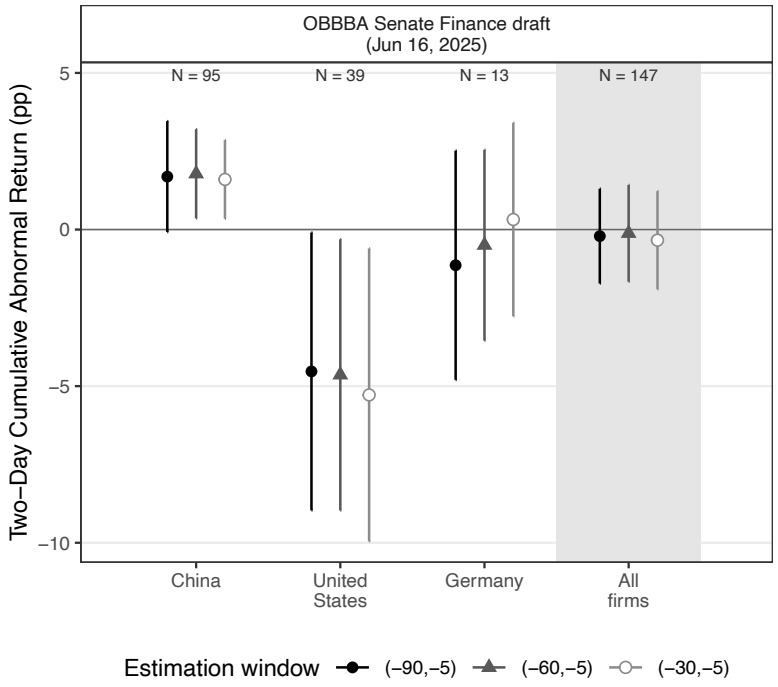


Figure 3: Abnormal Returns After OBBBA Draft Announcement

Note: Mean country-level abnormal returns with 95% confidence intervals using portfolio test standard errors. Firm-level abnormal returns obtained with a three-factor model.

the Senate Finance Committee's draft. In contrast to the initial IRA announcement, the results reveal a pronounced divergence between U.S. and Chinese green firms. U.S. firms experienced negative abnormal returns averaging approximately 4.5 percentage points, while Chinese firms registered positive abnormal returns of approximately 1.5 to 2 percentage points. German firms also experienced negative abnormal returns, although these estimates are again statistically indistinguishable from zero. The aggregate response across all firms is similarly close to zero, masking substantial differences in the distribution of gains and losses across countries. These findings are consistent with the logic of market diversion: the anticipated withdrawal of U.S. industrial support reduced the expected profitability of domestic producers while improving the competitive position of Chinese firms. The contrast with the initial IRA announcement suggests that the distributional consequences of preferential green industrial policy may become more apparent

as uncertainty about its implementation and competitive effects is resolved.[11]

Finally, we examine whether Chinese firms' responses to the OBBBA draft depend on their ability to benefit from U.S. production incentives. Appendix Figure A-2 compares abnormal returns among Chinese firms with and without manufacturing facilities in the United States. We gather the locations of manufacturing plants associated with the IRA from Clean Investment Monitor and match them to ownership by Chinese firms in our sample. Following the announcement of the OBBBA Senate draft, Chinese firms without U.S. manufacturing facilities experienced larger positive abnormal returns than those with U.S. operations. This pattern is consistent with the logic of market diversion: firms unable to benefit directly from U.S. manufacturing incentives stood to gain more from their withdrawal. Yet the difference is small in magnitude—less than one percentage point—and not statistically significant. The small number of Chinese firms with U.S. manufacturing facilities ($N = 6$) likely limits the precision of these estimates.

4.1 Robustness

To probe the robustness of our core results, we reestimate abnormal returns using longer estimation windows of $(-120, -5)$, $(-150, -5)$, and $(-180, -5)$ trading days, compared with our baseline window of $(-90, -5)$. Appendix Figures A-3 and A-6 present the results. Across all three policy announcements, the estimated abnormal returns remain remarkably stable in magnitude and direction. Following REPowerEU, Chinese firms continue to experience positive abnormal returns of approximately 3 percentage points, while the IRA announcement generates gains of approximately 13 percentage points among U.S. firms. Both effects remain statistically significant across estimation windows.

Similarly, following the OBBBA Senate draft announcement, U.S. firms experience losses of approximately 4.5 percentage points, while Chinese firms gain approximately

[11] For a similar perspective that the U.S. had reaped gains from the IRA, see https://www.theatlantic.com/ideas/2026/08/biden-climate-ira-2024/688179/

1.5 percentage points. Although the precision of these estimates declines somewhat with longer estimation windows, their magnitudes remain virtually unchanged. Longer estimation windows may introduce additional noise into the estimation of expected returns, particularly when they encompass periods of substantial market volatility or structural change. This concern is especially relevant for the OBBBA analysis, where the longer windows extend into the period of market disruption surrounding the Trump administration's April 2025 tariff announcements. Nevertheless, the stability of the point estimates suggests that our substantive findings are not sensitive to the choice of estimation window.

Our results are also similar when we weight firms by the inverse of their estimation window residual variance (Appendix Figures A-5 and A-8) and when we replace the factor model with a LASSO-weighted counterfactual constructed from the returns of other stocks traded in each firm's home market (Appendix Figures A-4 and A-7). Appendix Table A-12 compares LASSO-weighted counterfactual returns with those of our main specification. The factor model performs better out-of-sample than the LASSO: its median cross-validated R^2 exceeds the LASSO's for Chinese and U.S. firms in every event and estimation window but one.

We additionally report two further tests (Table A-14) that also account for cross-sectional correlation in how policy shocks affect firms across the globe: an empirical p-value, equal to the share of two-day windows in the estimation period whose cumulative average abnormal return is at least as large in absolute value as the event window estimate, and the standardized test of Kolari and Pynnönen (2010), which adjusts for the average pairwise correlation of firms' estimation window abnormal returns. Results are consistent with our main specification across these alternative tests.

5 Discussion

Our findings suggest that the international consequences of green industrial policy depend on the extent of preferential provisions to domestic producers. The REPowerEU plan, which expanded expected demand for clean energy without conditioning support on where equipment is made, raised the valuations of clean energy firms broadly, including Chinese manufacturers. The Inflation Reduction Act also expanded the market, and the average clean energy firm in our sample gained when its passage became likely. But those gains were concentrated among U.S. firms, whose valuations rose significantly, while Chinese and German firms saw no meaningful change. Investors appear to have expected the IRA to enlarge the market and to ensure that the additional value accrued largely to domestic producers.

The absence of a foreign response to the IRA admits two interpretations. The legislation may simply have been irrelevant to foreign firms. Alternatively, its expansionary and diversionary effects on foreign firms may have offset one another: Chinese manufacturers stood to gain from a larger U.S. and global market but to lose market share to subsidized U.S. competitors. The Senate Finance Committee's draft of the One Big Beautiful Bill Act allows us to distinguish between these possibilities. Following its release, U.S. clean energy firms lost value, Chinese firms gained, and the average valuation across all green firms was essentially unchanged. If the IRA had been irrelevant to foreign firms, rolling it back should have been irrelevant as well. Instead, the pattern suggests that investors expected retrenchment to shift market share from U.S. to Chinese producers, which implies that the IRA's null effect on foreign firms masked offsetting expansion and diversion. Preferential green industrial policies may thus strengthen domestic industries while weakening the competitive position of firms abroad, even when their net effect on foreign firms is not visible at the moment of adoption.

The firm-level evidence is consistent with this interpretation. Following REPowerEU,

Chinese firms with a high share of sales abroad gained more than Chinese firms focused on the domestic market, as one would expect if European demand expansion benefited exporters in particular. No such difference appears following the IRA, consistent with the view that its expansionary benefits did not reach foreign producers on net. Chinese firms are major producers of solar panels, batteries, and other technologies that compete directly with the U.S. manufacturers targeted by the IRA's production credits and domestic content provisions (Nahm 2021; Allan and Nahm 2025), so they are the foreign firms most exposed to the diversionary features of green industrial policy.

These findings have implications for how scholars understand the political economy of decarbonization. Existing research emphasizes how climate policy reshapes domestic coalitions by creating beneficiaries who subsequently support more ambitious climate action (Meckling et al. 2015; Colgan, Green and Hale 2021). Our results suggest that whether this logic extends beyond national borders depends on policy design. Demand-side policies without preferential provisions, such as REPowerEU, create beneficiaries abroad as well as at home: Chinese exporters gained from the expansion of European demand. Preferential policies such as the IRA concentrate those gains domestically, and the OBBBA evidence suggests that they do so in part at the expense of foreign competitors. Strategies that build domestic political support for climate policy by tying benefits to domestic production may therefore weaken the economic interests that could support decarbonization elsewhere. As governments increasingly use industrial policy not only to accelerate the energy transition but also to cultivate national clean energy industries, the politics of decarbonization may become increasingly shaped by international competition over the location of production, investment, and profits. Understanding this emerging political economy requires attention not only to how climate policy expands clean energy markets, but also to how it redistributes competitive advantage across them.

Several limitations qualify our conclusions. First, abnormal returns capture investors' expectations at the time of each announcement, not realized outcomes. Because parts

of each policy were anticipated, most clearly in the case of REPowerEU, whose contents circulated before its release (Bayer, Crippa and Genovese 2026), our estimates are best interpreted as lower bounds on the revisions in expected value that each policy produced. Second, our sample includes only publicly traded firms and omits private producers and unlisted segments of the clean energy economy, which may respond differently.

6 Conclusion

We compare the effects of two major clean energy policies announced in 2022: REPowerEU and the Inflation Reducation Act. The European Union's REPowerEU plan expanded expected demand for clean energy without favoring European producers, and investors responded by raising the valuations of clean energy firms broadly, including Chinese manufacturers. The agreement that led to the passage of the U.S. Inflation Reduction Act likely also expanded the market, but its gains accrued to U.S. firms, and foreign firms showed no statistically meaningful response. Evidence from the Senate's 2025 draft rollback of the IRA suggests that this null did not mean the IRA was irrelevant abroad: when its provisions were curtailed, U.S. firms lost value and Chinese firms gained. Investors appear to have viewed the IRA not only as a policy that expanded clean energy markets but also as one that diverted market share toward domestic producers.

These results underscore the importance of incorporating international spillovers into analyses of green industrial policy. Existing research has focused primarily on the domestic consequences of these policies, emphasizing their capacity to accelerate investment, create employment, and cultivate political constituencies that support decarbonization. Our findings suggest that the international consequences of these policies are not uniform. Policies that expand demand without conditioning support on the location of production can benefit firms abroad, while preferential policies can create winners and losers across national borders. For understanding international spillovers, then, what matters is not

only how much a government spends on clean energy, but whether that spending is tied to where, and by whom, clean energy technologies are made.

The distinction between market expansion and market diversion is not specific to green industrial policy. Subsidies for products like semiconductors or pharmaceuticals can likewise enlarge a market, shift production toward favored producers, or both, and the design we use here could be applied to those cases (Juhász, Lane and Rodrik 2024). What distinguishes the green case is the industrial policy's objective to—in addition to decarbonization—secure production at home, so diversion is largely the intended effect. Climate policy aims to reduce global emissions, and the benefits of doing so do not depend on where clean technologies are made. In clean energy markets, diversion therefore does more than transfer profits between countries' firms: if it raises costs or slows deployment abroad, it can work against the policy's own goal. For this reason, the balance between expansion and diversion matters more for green industrial policy than for industrial policy in most other sectors.

This distinction has implications for the political feedbacks emphasized in recent scholarship, which argues that industrial policy can strengthen support for climate action by creating firms, workers, and communities with a material stake in the expansion of clean energy industries (Meckling et al. 2015; Breetz, Mildenberger and Stokes 2018; Meckling 2021). When the gains from these policies are confined to domestic producers, coalition-building in one country may reshape political incentives elsewhere. Governments whose firms lose market share may respond with industrial policies of their own, and foreign firms acquire a stake in the retrenchment of policies that disadvantage them. Policies that expand demand without preferential provisions have the opposite effect: they give foreign firms a stake in another government's climate commitments, as Chinese exporters had in Europe's plan to accelerate solar deployment. The design choices that determine who gains from green industrial policy may therefore also determine which coalitions form across borders.

Whether these dynamics accelerate or impede decarbonization remains an open question. One possibility is that competition over clean energy industries encourages governments to expand support for low-carbon technologies in order to attract investment. Another is that it contributes to protectionism, fragmented supply chains, and competition over the location of green production rather than the pace of emissions reductions. The distinction between market expansion and market diversion offers one way to adjudicate between these possibilities, because the global climate benefits of industrial policy depend on the former while much of its political conflict stems from the latter. Answering this question will require evidence on realized investment and deployment rather than investors' expectations, and on a broader set of policies than the three announcements we study. As industrial policy becomes a central component of climate strategy, the future of decarbonization may depend not only on how much governments support clean energy industries at home, but also on how they design that support.

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